

A PREDICTIVE SYSTEM FOR IDENTIFYING BOT - GENERATED AD CLICKS

A PROJECT REPORT

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ABSTRACT

The Ad Click Fraud Detection Dashboard is an advanced system developed to identify and mitigate fraudulent activity in online advertising. Click fraud is a pervasive issue in digital marketing, where automated bots or malicious actors generate false clicks on advertisements. This not only leads to substantial financial losses for advertisers but also skews the analytics used to measure campaign effectiveness.

This project leverages machine learning models, specifically RandomForest and XGBoost classifiers, to predict whether an ad click originates from a genuine human user or a fraudulent source. The system is trained on historical click data with features such as IP address, application ID, device type, operating system, channel, and click timestamp. By analyzing these features, the models can accurately distinguish between human and bot clicks based on patterns and anomalies in user behavior.

The prediction models are integrated into a user-friendly web dashboard built with Flask, HTML, CSS, and JavaScript. The dashboard provides an interactive interface where users can input click-related data and obtain real-time predictions.

The primary objective of this system is to empower advertisers and businesses to detect suspicious clicks proactively, optimize their advertising budgets, and improve the accuracy of marketing analytics. This project demonstrates the practical application of machine learning for cybersecurity in digital advertising and provides a scalable foundation for future enhancements, such as real-time monitoring and integration with larger ad networks.

TABLE OF CONTENT

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	iv
	LIST OF FIGURE	ix
	LIST OF ABBREVIATIONS	x
1	INTRODUCTION	1
	1.1 OBJECTIVE	1
	1.2 OVERVIEW	1
	1.3 INTRODUCTION	2
2	LITERATURE SURVEY	4
3	SYSTEM ANALYSIS	14
	3.1 EXISTING SYSTEM	14
	3.1.1 DISADVANTAGES	14
	3.2 PROPOSED SYSTEM	14
	3.2.1 ADVANTAGES	15
4	REQUIREMENT SPECIFICATION	16
	4.1 HARDWARE REQUIREMENTS	16
	4.2 SOFTWARE REQUIREMENTS	16
	4.3 TECHNOLOGIES USED	17

4.3.1 PYTHON	17
4.3.1.1 INTRODUCTION TO PYTHON	17
4.3.1.2 WORKING OF PYTHON	18
4.3.1.3 ADVANTAGE OF PYTHON	18
4.3.1.4 ROLES OF PYTHON IN PREDICTION	19
4.3.1.5 PYTHON LIBRARIES USED	20
4.4 FLASK	20
4.4.1 WHAT IS FLASK?	20
4.4.2 ROLES OF FLASK IN PREDICTION	20
4.4.3ADVANTAGES OF FLASK FOR (UI)	21
4.4.4 HOW IT WORKS	22
4.4.5 WHERE FLASK IS USED	22
4.5 JUPYTER NOTEBOOK/ANACONDA	22
4.5.1 WHAT IS JUPYTER NOTEBOOK?	22
4.5.2 WHAT IS ANACONDA?	23
4.5.3 ROLE IN PREDICTION	23
4.5.4 ANACONDA USED FOR	23
4.5.5 WORKFLOW IN YOUR PREDICTION	24
4.5.6 ADVANTAGE OF JUPYTER NOTEBOOK	24
4.5.7 ADVANTAGE OF ANACONDA	24

	4.5.8 HOW THEY WORK TOGETHER	25
5	SYSTEM DESIGN SPECIFICATION	26
	5.1 ARCHITECTURE DIAGRAM	26
	5.2 WROKFLOW DIAGRAM	27
	5.3 USE CASE DIAGRAM	28
	5.4 CLASS DIAGRAM	29
	5.5 SEQUENCE DIAGRAM	30
6	SYSTEM DESIGN IMPLEMENTATION	31
	6.1 MODULE DESCRIPTION	31
	6.1.1 MODULES	31
	6.1.2 CLASSIFIER TRAINING	39
7	MACHINE LEARNING &ALGORITHM	
	TECHNIQUES	40
	7.1 APPLYING MACHINE LEARNING	40
	7.1.1 DECISION TREE	40
	7.1.2 PERFORMANCE MATRICES	41
	7.1.3 CONFUSION MATRIX	41
	7.1.4 ENSEMBLE AND PREDICTION MODULE	42

	7.1.5 EVALUATION MODULE	43
	7.1.6 DEPLOYMENT MODULE	43
	7.1.7 CONTINUOUS LEARNING MODULE	43
	7.2 ALGORITHM TECHNIQUES USED	44
8	CONCLUSION AND FUTURE ENHANCEMENT	45
	8.1 CONCLUSION	45
	8.2 FUTURE ENHANCEMENT	46
	APPENDIX 1	47
	APPENDIX 2	67
	REFERENCES	71

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
5.1	ARCHITECTURE DIAGRAM	26
5.2	WROKFLOW DIAGRAM	27
5.3	USE CASE DIAGRAM	28
5.4	CLASS DIAGRAM	29
5.6	SEQUENCE DIAGRAM	30

LIST OF ABBREVIATIONS

ABBREVIATIONS	DESCRIPTIONS
SVM	Support Vector Machine
DT	Decision Tree
RF	Random Forest
XGB	XGBoost
ML	Machine Learning
AI	Artificial Intelligence
DL	Deep Learning